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An exploration on the effects of varying mass on a harmonic oscillator's damping ratio

Introduction

When I learnt the concept of simple harmonic motion (SHM) in class, I didn't acknowledge that the content we were learning only applied to ideal situations. Visualising the sinusoidal curve, I expected the amplitude of motion to remain constant throughout the SHM until one day our teacher allowed us to experiment with mass-on-spring systems in class. When I extended the mass on the spring beyond the equilibrium point of the system and then released it, I noticed that the maximum amplitude which the mass reached was getting smaller with each cycle. As such, I realised that the amplitude of SHM decreases over time in the real world; after reading on the topic I understood this to be the result of damping: the effect that tends to reduce the amplitude of oscillations. Varying the mass on the system affected the rate of decrease in amplitude. As such, I wanted to explore:

What is the relationship between the mass attached to a mass-on-spring system and the system's damping ratio?

Background information

Simple harmonic motion is motion for which the acceleration of an object is directly proportional to its displacement from a fixed point and is always directed towards the fixed point. Ideally, in a vertical mass-on-spring system, the amplitude (maximum displacement) reached by the mass each time should be the same each time. However, in the real world, simple harmonic systems are subject to the effects of damping. Damping involves the reduction in the amplitude of vibrations. There are three main types of damped systems: critically damped, heavy damped, and underdamped (Massachusetts Institute of Technology, 2011). In a critically damped SHM system, the mass block returns to equilibrium quickly, with no oscillations whatsoever. Heavy damped systems are similar to critically damped systems in that there are no oscillations, but the mass takes far longer to return to the equilibrium position. In underdamped systems, however, oscillations occur, but the amplitude of oscillation decreases over time due to the effects of damping. In a vertical mass-on-spring system, this can arise from air resistance acting on the mass as it moves through the air, or fluid resistance if the experiment is carried out under water. In this experiment, I will be focusing on underdamped SHM systems. Figure 1 displays displacement-time graphs for a system with no damping and an underdamped SHM system (Author's Own, 2019):

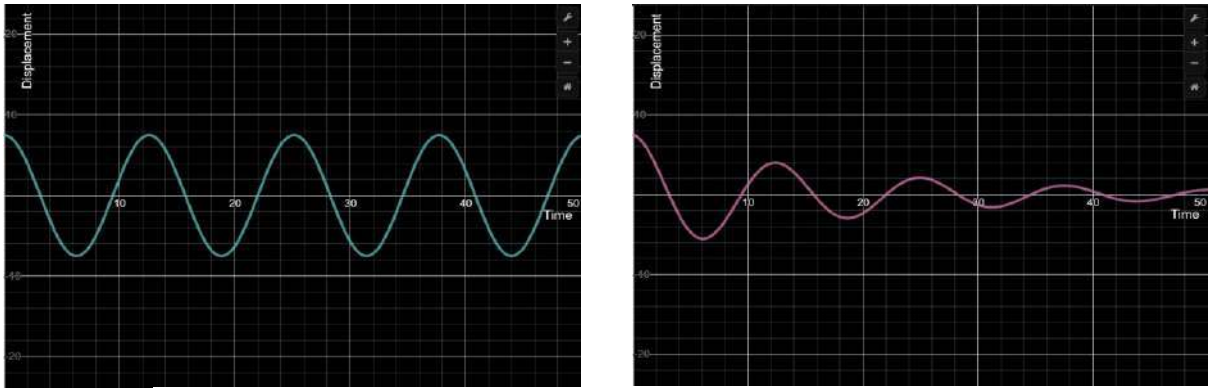


Figure 1: Graphical comparison of Undamped and damped SHM

For ideal, undamped SHM, the equation for the displacement-time graphs is given by:

$$x = A\cos(\omega t)$$

Where A and ω are constants representing amplitude and angular frequency of oscillation respectively

However, for damped SHM, the equation for the displacement-time graph is more complex:

$$x = Ae^{-t}\cos(\omega t)$$

Where A and ω are the same constants as above, and e is Euler's number. This modification to the undamped SHM equation causes the amplitudes of successive oscillations to decrease in value.

(Note: x is the symbol for displacement, and t is the symbol for time)

Both these equations assume the SHM starts from the maximum positive displacement of the system.

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The aim of this experiment is to determine the relationship between the mass attached to the spring and the damping ratio of a mass-on-spring system. Once this aim is complete, we will also be able to predict our value for the viscous damping coefficient of the SHM system that I set up. The viscous damping coefficient of a system is a theoretical parameter that encapsulates the causes of damping on a SHM system. It cannot be measured using laboratory equipment. Its value varies for a given damping system, and is dependent on a number of factors including the density of the medium, the drag force on the system, the velocity of the system, and the surface area of the head of the oscillator. Damping ratio, not to be confused with the damping coefficient, is a dimensionless measure of how the oscillations in a simple harmonic system decay due to disturbance (damping). Equation 1 below links damping ratio and mass (Brown University, n.d.):

$$\zeta = \frac{c}{2\sqrt{mk}}$$

Equation 1

$\zeta = damping\ ratio\ \ c = viscous\ damping\ coefficient\ \ m = mass\ on\ spring\ \ k = Hooke's\ constant$

Given this equation, it is evident that varying the mass has an effect on the damping ratio. However, there isn't a direct way of measuring damping ratio, so we must use an indirect method to determine values.

For damped simple harmonic motion equation, the logarithmic decrement is defined as the natural logarithm of the ratio of the amplitudes of any two peaks (On the damped SHM graph) (Vandiver, 2011):

$$\delta = \frac{1}{n}\ln\left(\frac{x(t)}{x(t+nT)}\right)$$

Equation 2

$\delta = log\ decrement\ \ n = no.of\ cycles\ \ x(t) = initial\ amplitude\ \ x(t+nT) = amplitude\ after\ n\ cycles$

For each trial involving a change in mass, we will need to calculate the logarithmic decrement for the motion graphs obtained. This value for the logarithmic decrement is then used in the calculation of damping ratio. The equation linking damping ratio to logarithmic decrement is as follows (Casiano, 2016):

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

Equation 3

$\zeta = damping\ ratio\ \ \delta = logarithmic\ decrement$

Using equations 2 and 3, and motion graphs that are obtained, we can determine the damping ratio of the mass-on-spring system at each mass. Plotting values of damping ratio against mass would then allow us to determine the relationship between the two variables.

Hypothesis

When the mass attached to the system is increased, the value of denominator increases, which results in a decrease in the value of the damping ratio.

We have the following equation linking the damping ratio and the mass attached to a mass-on-spring system:

$$\zeta = \frac{c}{2\sqrt{mk}}$$

The inversely proportional relationship between damping ratio and the square root of mass, shown by this equation, allows us to propose the aforementioned hypothesis.

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Experimental Design and Method

Independent Variable (IV): Mass attached to mass-on-spring system

The IV is changed by adding slotted masses to a mass hook attached to the spring. This allows us to change the mass on the system in fixed increments, and explore the effect of changing mass on the DV

Dependent Variable (DV): Damping ratio

A Vernier motion sensor is used in the plotting of SHM graphs for each mass using LoggerPro software. Using these graphs in conjunction with equation 2 allows us to calculate the logarithmic decrement, before these values are used in calculation of the respective damping ratios, using equation 3.

Control Variables:

Variable	Significance	How to control?
Nature of spring	The damping ratio for different springs is dependent on the value of the Hooke's constant, which varies for each spring. Keeping the dimensions of the spring the same ensured that mass on the mass-on-spring system was the only variable being changed.	Dimensions including length (0.0650 ± 0.0001 m), diameter (0.0150 ± 0.0001 m), and material were constant (steel). The same spring was used for all trials. The Hooke's constant that I determined was 0.06331 ± 0.00046 Nm^{-1}.
Initial displacement of mass	Displacing the mass from equilibrium position, transfers chemical energy to elastic energy in the spring, which causes oscillations. The amount of energy transferred depends on the displacement from equilibrium. Keeping initial displacement constant means that all trials start with same potential energy.	For each trial, the mass was displaced upwards by 0.025m to start the experiment. This will be done using the help of a metre ruler.
Number of cycles considered	Depending on the mass attached to the mass-on-spring system, the damping ratios are expected to vary. Given this information, we know that the maximum amplitude of systems with different mass will decrease at different rates. As such, to keep the calculations constant, we calculate the damping ratio after a fixed number of cycles of oscillation	Amplitude was analysed after 45 cycles for each trial
Medium in which the experiment was carried out	Different mediums offer varying resistance to SHM systems based on their density. Given that we have a viscous damping coefficient in our central equation 1, the coefficient needs to be constant.	All trials were carried out in the same part of the my physics lab in school. The air around the system was the same for each trial
Surface Area perpendicular to oscillation direction	At the bottom of the mass hanger, I attached a piece of card. To ensure that viscous damping coefficient of my SHM system was constant, this area had to be kept constant, as the value of the coefficient is dependent on the surface area perpendicular to the direction of oscillation.	A lightweight 0.1m by 0.1m square card was attached to the bottom of the mass hanger. This ensured that the surface area of the oscillator colliding directly with the air particles below it was constant

Equipment

- Vernier motion sensor x1
- Vernier Data logger x1
- Piece of card x1(that can be detected by motion sensor)
- 100g mass hanger x1
- 100g masses x6
- Spring x1(see specifications above)
- Retort stand x2
- Meter ruler x1
- G-clamp x2

Safety considerations

- To prevent retort stands falling, using a G-clamp may be useful in securing the stand in position
- The mass hanger may unhook itself from the spring while in motion. Falling masses can injure feet, and can damage sensors. Using a cage to protect the motion sensor may be useful. Lightweight tape could be used to secure the hanger, while ensuring that it is still possible to place or remove masses

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Procedure

Procedure for measurement of Hooke's constant (Preliminary experiment)

Hooke's Law states that the extension of a spring is directly proportional to the force applied to it. This is summarized in the following equation:

$$F = -kx$$

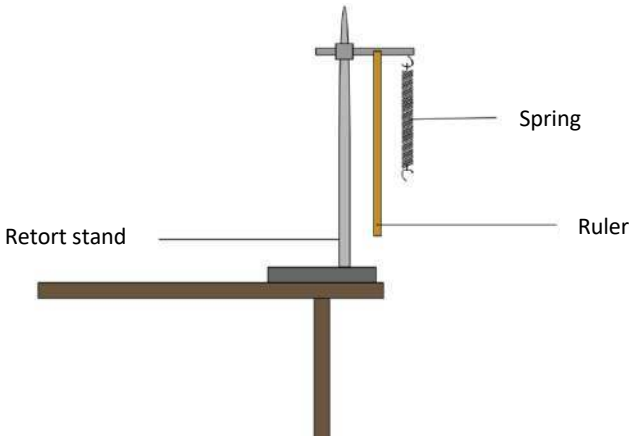
Equation 4

$$F = Force \quad k = Hooke's\ constant \quad x = displacement\ from\ equilibrium$$

The manufacturer of the spring I used in my experiment did not specify the Hooke's constant for the spring, so I had to carry out a separate experiment to do so myself:

1) Set up the apparatus as shown in Figure 2 (Author's Own, 2020):

Figure 2: Setup to determine the Hooke's constant of a spring



2) Hang a 100g mass hanger from the bottom hook of the spring
3) Measure the extension of the spring with the help of a meter ruler thrice
4) Slot a 100g mass onto the hanger, and repeat step 3
5) Slot 2, 3, 4, and 5 100g masses onto the hanger (in separate experiments), and repeat step 3

Table 1: Extension of spring at different masses

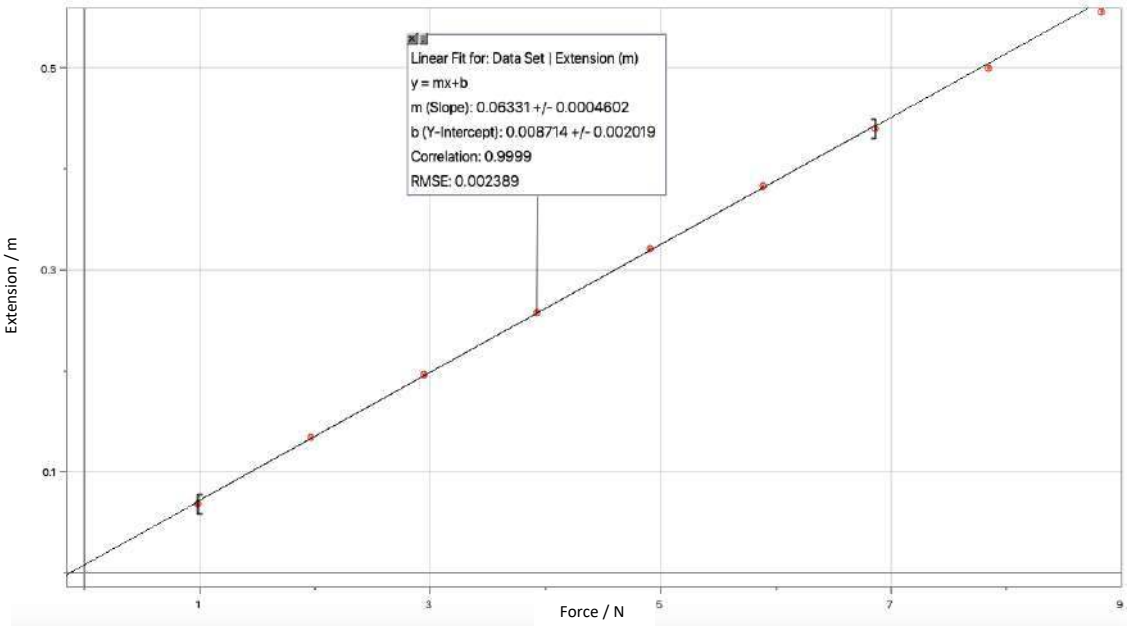
Mass (m)/ ±0.001kg	Force (F)/ ±0.00981 N	Length (l)					Extension (x)	Abs. Unc in Extension
		Trial 1	Trial 2	Trial 3	Average	Abs Unc. In average		
0.100	0.981	0.1330	0.1320	0.1340	0.1330	0.0010	0.068	0.002
0.200	1.962	0.1980	0.1990	0.1990	0.1990	0.0005	0.134	0.001
0.300	2.943	0.2620	0.2600	0.2610	0.2610	0.0010	0.196	0.002
0.400	3.924	0.3240	0.3230	0.3220	0.3230	0.0010	0.258	0.002
0.500	4.905	0.3860	0.3860	0.3860	0.3860	0.0005	0.321	0.001
0.600	5.886	0.4470	0.4480	0.4480	0.4480	0.0005	0.383	0.001
0.700	6.867	0.5050	0.5060	0.5050	0.5050	0.0005	0.440	0.001
0.800	7.848	0.5650	0.5640	0.5650	0.5650	0.0005	0.500	0.001
0.900	8.829	0.6200	0.6210	0.6210	0.6210	0.0005	0.556	0.001

Sample calculation for 0.100kg

Calculation	Uncertainty Propagation
$W = mg = 0.1kg \times 9.81\frac{N}{kg} = 0.981N$	$0.001kg \times 9.81\frac{N}{kg} = \pm 0.00981N$
$Average\ length = \frac{T1+T2+T3}{3} = \frac{0.1330+0.1320+0.1340}{3}$ $= 0.1330m\ (3.s.f)$	$\Delta Avg. = \frac{l_{max} - l_{min}}{2} = \frac{0.1340-0.1320}{2} = \pm 0.0010m$
$Extention = Avg.\ Length - 0.065 = 0.1330 - 0.065$ $= 0.068m$	$\Delta x = 0.0010 + 0.0010 = \pm 0.002$

In order to determine the Hooke's constant for the spring, I plotted a graph of Force against Extension. The gradient of this graph should be numerically equal to the Hooke's constant for my spring. This graph is shown in Figure 3 (Note: uncertainties are too small in comparison relative to the scale, and are difficult to see on the graph):

Figure 3: Graph of Extension against Force

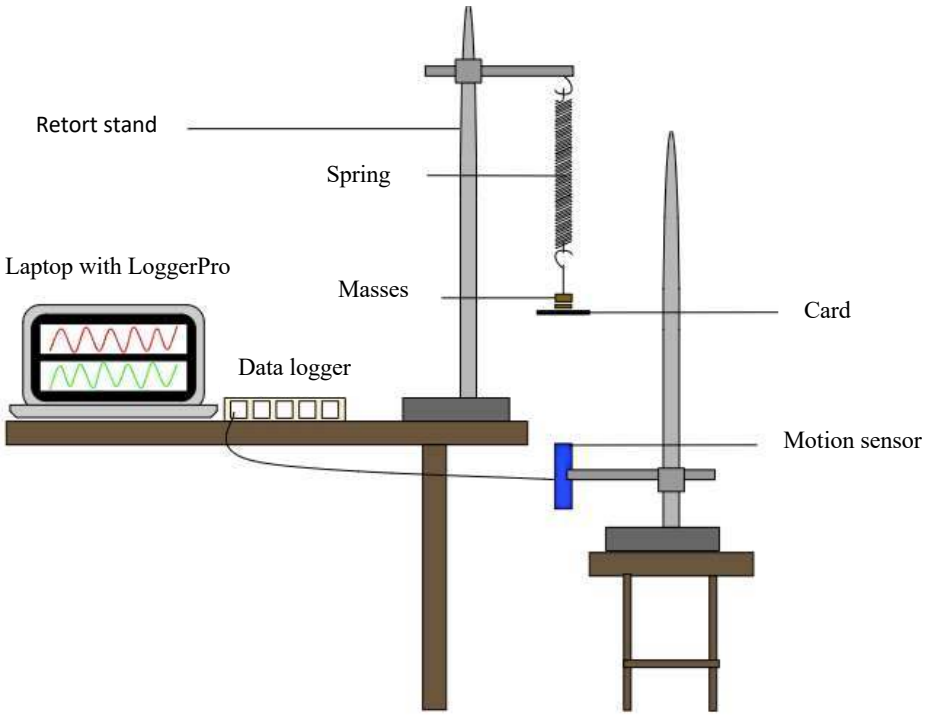


As such, the Hooke's constant of the spring I used in my experiment was 0.06331± 0.00046 N/m
This value will be required during the analysis of our results later in this exploration.

Procedure to determine relationship between mass and damping ratio

1) Set up apparatus as shown in Figure 4 and add a 100g mass block to the 100g mass hanger. (I chose to avoid taking measurements for the 100g mass hanger by itself, as it experienced large amounts of swaying motion during preliminary experiments)(Author's Own, 2020):

Figure 4: Experimental setup to determine relationship between mass and damping ratio



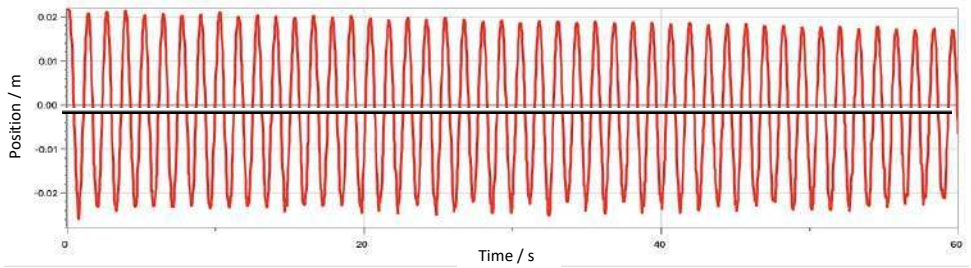
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2) Set up the software such that data collection occurs for 60s at the maximum sampling rate of the sensor
3) Calibrate the motion sensor (Toolbar > Experiment > Zero)
4) Push the bottom of the mass hanger up by 0.025m, press "play" on LoggerPro, and release the oscillator
5) Allow the system to oscillate. LoggerPro software should automatically stop data collection after 60s
6) Determine the amplitude $x(t + nT)$ of a crest after 45 cycles of oscillations
7) Repeat steps 1-6 four more times and take an average value for $x(t + nT)$
8) Determine the logarithmic decrement for the mass-on-spring system at 200g using equation 2, the constant maximum initial displacement $x(t)$, and the value calculated in step 7
9) Using the value found for the decrement in step 8, find the value of the damping ratio using equation 3
10) Repeat steps 1-9 for total masses of 300g, 400g, 500g, 600g, 700g, 800g and 900g ("Total mass" includes the mass of the hanger. As mass is increased, the spring undergoes extension, so the position of the sensor must be adjusted using the retort stand clamp, ensuring that the device is between 0.15-0.25m below the card for optimal collection. The distance between the sensor and card does not impact results)

Results

Figure 5 displays a sample SHM graph obtained in Trial 1 with a total mass of 200g, starting from a displacement of 0.025m. Damping is visible, evidenced by the decrease in amplitude:

Figure 5: Sample SHM graph (200g, Trial 1)



On many graphs, the sensor didn't detect the equilibrium position as 0, even though the sensor was calibrated before each trial. As such, we assume the equilibrium position to be the mean displacement each time. We can make this assumption, as the equilibrium position on the graph, even though it is non-zero, remains around the same for the duration of the trial. I obtained such SHM graphs for 5 trials at each mass between 200g-900g. These were then analysed to determine the average amplitude after 45 cycles for each mass attached to the oscillator:

Table 2: Raw data (amplitudes of oscillation after 45 oscillations at each mass)						
Mass / ±0.001kg	Amplitude after 45 of cycles $(x(t + nT), where\ n\ is\ now\ 45)$ / ±0.00001 m					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average
0.200	0.01949	0.01893	0.01838	0.01875	0.01884	0.01888
0.300	0.01972	0.01913	0.02002	0.02083	0.01970	0.01988
0.400	0.02109	0.02029	0.01979	0.02025	0.02120	0.02052
0.500	0.02001	0.02148	0.02160	0.02087	0.02103	0.02100
0.600	0.02150	0.02211	0.02159	0.02111	0.02124	0.02151
0.700	0.02186	0.02089	0.02201	0.02196	0.02148	0.02164
0.800	0.02288	0.02089	0.02220	0.02097	0.02225	0.02184
0.900	0.02130	0.02095	0.02300	0.02340	0.02090	0.02191

Note: Uncertainty in mass is assumed to be the smallest significant digit stated by the manufacturer

Sample calculation of Average amplitude for 0.200kg

Calculation	Uncertainty Propagation
$Average\ amplitude\ after\ 45\ cycles = \frac{T1+T2+T3+T4+T5}{5}$ $= \frac{0.01949+0.01893+0.01838+0.01875+0.01884}{5}$ $= 0.01888m\,(5.d.p)$	Uncertainty in average amplitude assumed to be the same as the trials (±0.00001m). Consideration of max-min calculation is in the next section, when calculating uncertainty in δ.

Qualitative observations

- The mass-on-spring system often experienced a pendulum like motion while during its oscillation. This swinging motion may have impacted the results

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Processed data

Next, I determined the logarithmic decrement and damping ratio for each mass using equations 2 and 3 respectively:

Table 3: Logarithmic Decrements and Damping Ratios of SHM systems holding different masses

Mass / ±0.001kg	Avg. amplitude after 45 of cycles / ±0.00001m	Logarithmic Decrement (δ)	Uncertainty in (δ)	Damping Ratio (ζ)	Uncertainty in (ζ)
0.200	0.01888	0.00624	0.00065	0.00099	0.00010
0.300	0.01988	0.00510	0.00095	0.00081	0.00015
0.400	0.02052	0.00439	0.00076	0.00070	0.00012
0.500	0.02100	0.00388	0.00085	0.00062	0.00014
0.600	0.02151	0.00334	0.00051	0.00053	0.00008
0.700	0.02164	0.00321	0.00058	0.00051	0.00009
0.800	0.02184	0.00301	0.00097	0.00048	0.00015
0.900	0.02191	0.00281	0.00126	0.00045	0.00020

Sample calculations for logarithmic decrement and damping ratio at 200g

Calculation

In order to determine our logarithmic decrement, we make use of equation 2:

$$\delta = \frac{1}{n}\ln\left(\frac{x(t)}{x(t+nT)}\right)$$
$$= \frac{1}{45}\ln\left(\frac{0.025}{0.0188}\right) = 0.00624\,(5.d.p)$$

While making this calculation, it is important to consider that the initial displacement was not always a constant 0.025m, even though this is one of the control variables in the experiment. This was partly due to the difficulty in ensuring that the spring was displaced vertically by exactly 0.025m. As such, there is a deviation in the initial amplitudes for different trials, which contributes to random error in the experiment. Given our access to the 40 trials in the this experiment, we can calculate the standard deviation in our initial displacement:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$
$$= \sqrt{\frac{(0.0176 - 0.0256)^2 + \ldots + (0.0277 - 0.0256)^2}{40}}$$
$$= 0.0037233\ldots \approx \pm 0.004m$$

Because our mean value for the initial displacement is relatively close to our desired 0.025m, we will make the assumption that initial displacements for all trials was 0.025. However, the existence of the standard deviation may contribute to random error, which is discussed further on in this exploration

Uncertainty Propagation

The uncertainty propagation for logarithmic calculations is extensive, and far beyond the reach of the Physics HL course, so I determine the uncertainty in the logarithmic decrement using the given formula:

$$\Delta\delta = \frac{\delta_{max} - \delta_{min}}{2}$$
$$\Delta\delta = \frac{\frac{1}{n}\ln\left(\frac{x(t)}{x(t+nT)_{min}}\right) - \frac{1}{n}\ln\left(\frac{x(t)}{x(t+nT)_{max}}\right)}{2}$$
$$\Delta\delta = \frac{\frac{1}{45}\ln\left(\frac{0.02500}{0.01838}\right) - \frac{1}{45}\ln\left(\frac{0.02500}{0.01949}\right)}{2}$$
$$\Delta\delta = \frac{0.00684 - 0.00553}{2} = \pm 0.00065\,(5.d.p)$$

A larger value of the amplitude after 45 oscillations, gives a smaller logarithmic decrement. As such, it is appears to be (min – max) in the calculation

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To determine the damping ratio, we use the value of logarithmic decrement that we calculated, and use it in equation 3:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$
$$= \frac{0.00624}{\sqrt{4\pi^2 + 0.00624^2}} = 0.00099\,(5.d.p)$$

To propagate uncertainties for the damping ratio, we use the following properties:

$$if\ y = \frac{a}{b}\ then, \frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} \qquad y = a^n\ then, \frac{\Delta y}{y} = \left| n\frac{\Delta a}{a} \right|$$

Given that the $4\pi^2$ inside the square root is a constant, uncertainty propagation of the denominator leaves us with:

$$\frac{\Delta\zeta}{\zeta} = \frac{\Delta\delta}{\delta} + \left[\frac{1}{2} \times \left\{ \frac{0 + \left(2 \times \frac{\Delta\delta}{\delta} \times \delta^2 \right)}{4\pi^2 + \delta^2} \right\} \right]$$

When simplified:

$$\frac{\Delta\zeta}{\zeta} = \frac{\Delta\delta}{\delta} + \frac{\delta \cdot \Delta\delta}{4\pi^2 + \delta^2}$$
$$\frac{\Delta\zeta}{\zeta} = \frac{0.00065}{0.00624} + \frac{(0.00624 \cdot 0.00065)}{4\pi^2 + 0.00624^2}$$
$$\frac{\Delta\zeta}{\zeta} = 0.10417$$
$$\Delta\zeta = 0.10417 \times \zeta = 0.10417 \times 0.00099 = \pm 0.00010$$

Graph

The data needs to be linearized so that we can determine the relationship between mass and damping ratio.

$$\zeta = \frac{c}{2\sqrt{mk}}$$

Given equation 1, which is shown above, I expected to obtain a linear graph if I plotted damping ratio against the reciprocal of the square root of mass. This is because the equation follows the structure of linear equation y = mx, where in our case damping ratio is y, $\frac{1}{\sqrt{m}}$ is x, and $\frac{c}{2\sqrt{k}}$ is our value for m, the gradient of the linearized graph. Additionally, given the nature of this equation, ideally there should not be a y-intercept for my final plot. The table below displays masses, their corresponding values of $\frac{1}{\sqrt{m}}$ and damping ratios:

Table 4: Final processed data for graphical analysis

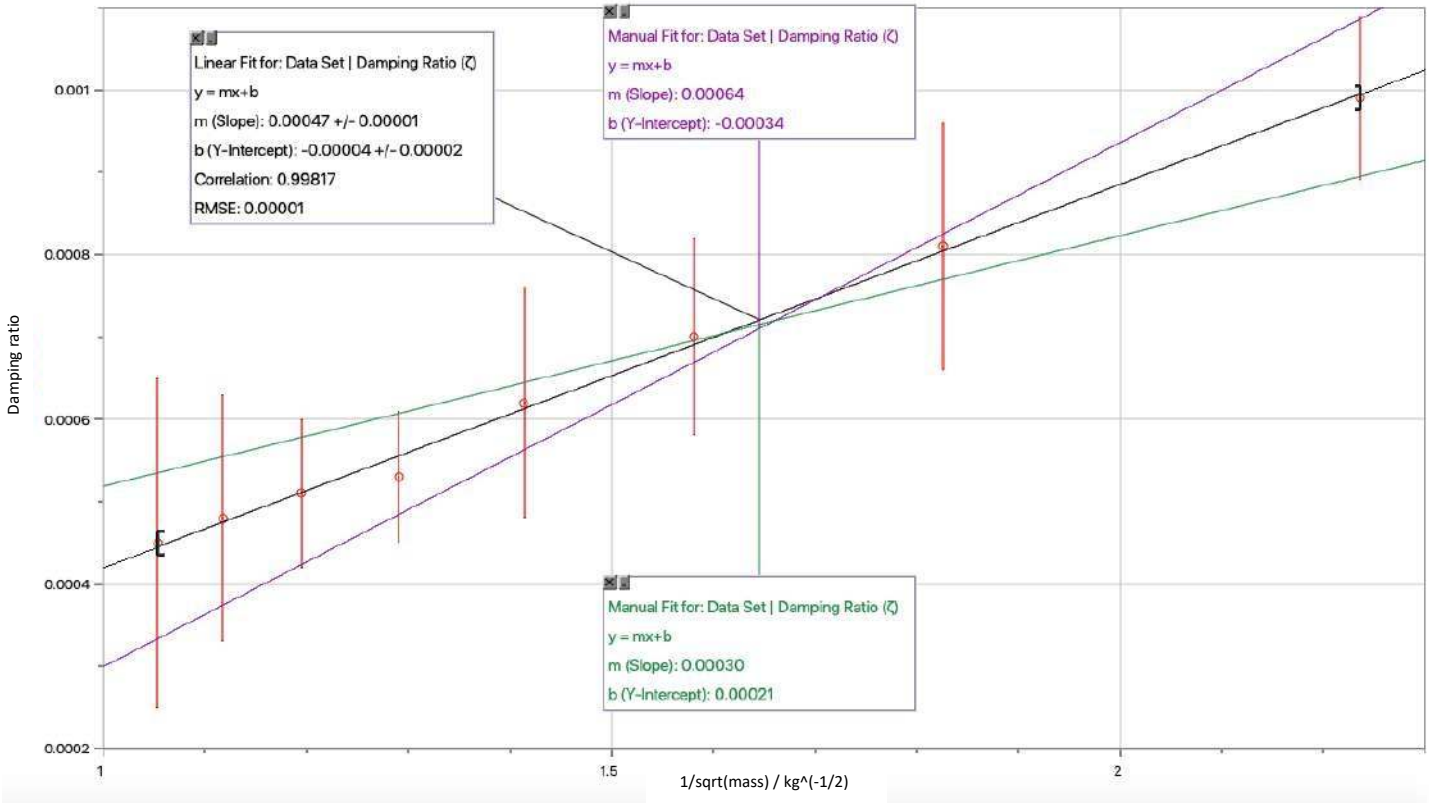
Mass / ±0.001kg	$\frac{1}{\sqrt{mass}}$ / kg^{-1}	Abs. Unc. in $\frac{1}{\sqrt{mass}}$ / kg^{-1}	Damping Ratio (ζ)	Uncertainty in (ζ)
0.200	2.23607	0.00056	0.00099	0.00010
0.300	1.82574	0.00030	0.00081	0.00015
0.400	1.58114	0.00020	0.00070	0.00012
0.500	1.41421	0.00014	0.00062	0.00014
0.600	1.29099	0.00011	0.00053	0.00008
0.700	1.19523	0.00009	0.00051	0.00009
0.800	1.11803	0.00007	0.00048	0.00015
0.900	1.05409	0.00006	0.00045	0.00020

Sample calculations for 200g

Calculation	Uncertainty Propagation
$\frac{1}{\sqrt{mass}} = \frac{1}{\sqrt{0.2}} = 2.23607$ Values in this table are to 5 decimal places were used, as the first significant figure of the smallest calculated uncertainty for the damping ratio was only at the 5th decimal place (Uncertainty for 0.600kg was 0.00008)	$given\ that\ if\ y = a^n\ then, \frac{\Delta y}{y} = \left| n\frac{\Delta a}{a} \right|$ $\frac{\Delta\frac{1}{\sqrt{mass}}}{\frac{1}{\sqrt{mass}}} = \left| -\frac{1}{2} \times \frac{\Delta mass}{mass} \right| = \frac{1}{2} \times \frac{0.001}{0.200}$ $\Delta\frac{1}{\sqrt{mass}} = 0.0025 \times \frac{1}{\sqrt{mass}} = 0.0025 \times 2.23607 = \pm 0.00559 kg^{-\frac{1}{2}}$

We can now plot damping ratio against $\frac{1}{\sqrt{m}}$ to obtain a linear plot. This plot is shown in Figure 6. It includes a trend line (black), a maximum line (purple) and a minimum line (green):

Figure 6: Plot of damping ratios against the square root mass



The gradient of the trendline is given as 0.00047, whereas the gradients of the maximum line and minimum line are 0.00064 and 0.00030 respectively. Using the same formula for uncertainties used before;

$$\Delta gradient = \frac{gradient_{\max} - gradient_{\min}}{2}$$

$$= \frac{0.00064 - 0.00030}{2} = \pm 0.00017 N^{-\frac{1}{2}} m^{\frac{1}{2}}$$

LoggerPro states the uncertainty in the gradient as ± 0.00001. While this is a more accurate value due to LoggerPro's computational abilities, using the steepest/flattest lines method is acceptable because the gradient uncertainty calculation by LoggerPro doesn't take our error bars into consideration. It only considers how close the data points are to the trend line. The gradient is used to calculate the damping coefficient for our system:

$$gradient = \frac{c}{2\sqrt{k}}$$

$$c = 2 \times gradient \times \sqrt{k}$$

(No units, as c is an arbitrary value)

$$c = 2 \times 0.00047 \times \sqrt{0.06331} = 0.00024 \ (5.d.p)$$

Using the same equation, the maximum and minimum values of c were calculated using the maximum and minimum values for the gradient. The maximum value of c was 0.00032, and the minimum value was 0.00015. Therefore, the uncertainty in the value of our viscous damping coefficient is:

$$\Delta c = \frac{c_{\max} - c_{\min}}{2}$$

(No units, as c is an arbitrary value)

$$= \frac{0.00032 - 0.00015}{2} = \pm 0.00009$$

Another way to determine the uncertainty in "c" is through the use of uncertainties in c and k:

$$\frac{\Delta c}{c} = \frac{\Delta gradient}{gradient} + \frac{\Delta k}{k}$$

(No units, as c is an arbitrary value)

$$\Delta c = \left(\frac{0.00017}{0.00047} + \frac{0.00046}{0.06331} \right) \times 0.00024 = \pm 0.00009$$

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Both calculations leave us with the same uncertainty so this value for the viscous damping coefficient of my system can be written as: **0.00024 ± 0.00009**. When expressed as a percentage uncertainty, the uncertainty in the calculated value for the viscous damping coefficient is **37.5%**.

While the value stated by LoggerPro for the y-intercept for the trendline is relatively small at -0.00047, the y-intercepts of the maximum and minimum lines are -0.00034 and 0.00021 respectively. This could mean there was small systematic error in the experiment, however, the origin falls between the intercepts of the maximum and minimum lines, so the expected trend does seem to have appeared. This is discussed in the conclusion.

Conclusion

From our results, the conclusion can be made that the trend predicted in the hypothesis does exist between the mass attached to a mass-on-spring system and the system's damping ratio; as the mass attached to a mass-on-spring system is increased, the damping ratio of the system decreases. However, while there is a linear trend shown between damping ratio and $\frac{1}{\sqrt{m}}$, the trendline obtained does not pass through the origin. This being said, the origin point (0,0) does lie within the y-intercept range between the maximum and minimum lines, so we can still claim that **there is directly proportional relationship between damping ratio and $\frac{1}{\sqrt{m}}$ as hypothesised.**

Given that the trendline was expected to pass through the origin, the existence of a y-intercept suggests that there was slight systematic error present in the experiment. Also, the size of error bars in relative to the scale are large, indicating significant random error, deeming the experiment imprecise. The source for this uncertainty and error in my experiment could either be: the swinging of the oscillator similar to a torsional pendulum; or perhaps the difficulty in precisely displacing the oscillator from an initial amplitude of 0.025m. Additionally, using a denser medium, such as water, might have made it easier see greater differences in damping ratios when the mass was changed. Overall, while the expected trend has been seen, I am only partially confident in the result of my experiment due the relatively large size of the error bars.

The correlation, or R-squared of my line of best fit is stated to be 0.99817, and my RMSE value is 0.00001 (as calculated by LoggerPro). These values suggest that there is a strong correlation between the data and the trend line. As such, my results are precise, while they are not necessarily fully accurate.

Given that the viscous damping coefficient is unique to a simple harmonic oscillator, and is too extensive to determine through other methods at my level, there is no literature value for the viscous damping coefficient for my value to compared with. However, if more time was allotted to exploring the functions of a SHM system, I would explore other ways of finding the viscous damping coefficient to be able to make a comparison.

Evaluation

Limitations

Method		
Limitation	**Significance**	**Solution**
Low number of repeats at each mass	Given the fluctuations in the raw data, which are emphasised by the size of the calculated standard deviation, the question the reliability arises	Due to the time constraints of the experimentation period and the rigour of the setup, it was difficult to carry out a desirable number of repeats. Completing 15 trials at each mass in would offer more reliable results
There was subtle swinging of the oscillator in a motion similar to that of a pendulum	Undesirable swaying of the oscillator may be a cause for random error in the experiment. This, in turn, affects the precision of the results.	Using a stiffer spring might reduce such motion. The spring in my experiment was relatively light, which meant that minute horizontal displacement could set off the pendulum motion. Stiffer springs act quickly to return to their original shapes, so such motion can be reduced. However, too stiff springs might also lead to swinging motion. So it is up to the experimenter to choose the correct spring Additionally, hanging the spring from a piece of string may help in reducing this unwanted motion (Institute of Physics, n.d.). This technique was tried, but I was unable to get the system to function appropriately. In future experiments this technique could be perfected to offer better results.

Clastify

Equipment		
Limitation	**Significance**	**Solution**
Difficulty in displacing the mass while ensuring that the mass is being moved vertically upwards	I used one hand to release the mass from its starting position, and the other to start data collection. Given that my hand was quite unstable, and my line of sight was normal to the plane of oscillation of the spring, it was difficult to ensure that the mass was being moved vertically upwards. This may have been a cause for the pendulum like motion discussed earlier	Provided that the pendulum like motion is reduced as a result of the solution discussed above, placing a column made of plastic or glass around the spring may aid the experimenter in ensuring that the mass is only displaced vertically upwards. If the pendulum like motion is reduced, the mass should not be striking the sides of this column, and so the column would have no effect on the damping of the system. I also realised, upon discussion with my supervisor, that I didn't necessarily need to release the mass at the same time as press play. Pressing play on LoggerPro first, and then carefully starting the oscillation would reduce the chances of displacing the mass incorrectly
The set up for the apparatus was time consuming.	Due to an inefficient work system, it was difficult to obtain a desirable number of repeats per mass. This reduces the reliability of my results.	Perhaps using the help of peers would make the set up faster, which would allow completion of more trials in less time.

Strengths

Method	
Strength	Significance
Use of Technology (LoggerPro)	In most cases, using technology to make measurements ensures more accurate data collection. While considering possible methods to take recordings, I came across the options of manual video analysis (analysing a video by eye) and using a motion sensor connected to LoggerPro. The second option meant that data would be automatically collected by the software. This allowed me to choose this option for data collection.
Clamping the motion sensor on a retort stand	Given that the distance of the motion sensor from the oscillator has no impact on the data collection, adjusting the sensor's position to a distance from the oscillator produced smoother SHM graphs. The sensor was clamped to a retort stand directly below the oscillator. The height at which the clamp was attached could be adjusted as required.
Equipment	
Strength	Significance
Use of Vernier Motion Sensor	Similar to the argument for using technology over manual video analysis, the use of a motion sensor ensured more accurate data collection
Use of a wrench to tighten retort stands	Tightening the retort stands ensured that any wobbling of the set up did not contribute to random error in the experiment
Use of lightweight masking tape to hold individual mass blocks together, and securing mass hanger to the spring	During my preliminary experiments, individual mass blocks jumped up, causing the spring to jerk. This led to significant fluctuations in amplitude readings. I even damaged a motion sensor as the mass hanger slipped off the spring, and landed on the face of the sensor. As such, using tape to secure all the mass together, not only improved the accuracy of measurements, but also ensured that the experiment was safe

Clastify

Extension

As an extension, it would be interesting to explore the effect of density on the damping ratio of a SHM system. Error comparison for experiments in different mediums would allow a clearer understanding of the effects of mass on the damping ratio of a SHM system.

Additionally, the viscous damping coefficient for the system could be determined through other methods. This would allow a comparison of values for the coefficient from a range of procedures, and find the most accurate value, which can be compared to this experiment. One method found was through the use of the drag force equation (Benson, 2014):

$$D = \frac{1}{2} c \rho A v^2$$

Equation 5

$D = Drag\ force\ \ c = viscous\ damping\ coefficient\ \ \rho = density\ (of\ medium)$

$A = Surface\ area\ of\ the\ bottom\ of\ the\ oscillator\ \ v = velocity\ of\ the\ oscillator$

If all the variables in the equation other than the viscous damping coefficient are known, then rearranging the equation would allow us to find a numerical value. This could be compared with the value for the viscous damping coefficient that I determined in this experiment.

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